

Classical Baselines and Quantum Amplitude Estimation for Option Pricing: Black–Scholes and Heston

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GitHub repository

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1 Risk-neutral pricing framework

1.1 Objective

Most pricing problems in quantitative finance reduce to computing a *discounted expectation*. Let Φ be a payoff paid at maturity T (it may depend on S_T or on the whole path $(S_t)_{t \in [0, T]}$). We denote the time-0 price by V_0 .

1.2 No-arbitrage and risk-neutral valuation

Under no-arbitrage, there exists a risk-neutral measure $\mathbb{Q}$ such that, using the money-market account

$$B_t = e^{rt} \quad (\text{or } B_t = \exp(\int_0^t r_u du) \text{ if rates are time-dependent}),$$

the discounted prices of traded assets are martingales under $\mathbb{Q}$. In particular, for a European-style payoff Φ paid at time T ,

$$V_0 = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_u du} \Phi \right]. \quad (1)$$

When r is constant this simplifies to

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\Phi]. \quad (2)$$

If the underlying pays a continuous dividend yield (or carry) q , it affects the dynamics under $\mathbb{Q}$ through a drift adjustment $(r - q)$.

1.3 Benchmark product: vanilla call/put

For maturity T and strike K , the vanilla payoffs are

$$\Phi_{\text{call}}(S_T) = (S_T - K)^+, \quad \Phi_{\text{put}}(S_T) = (K - S_T)^+.$$

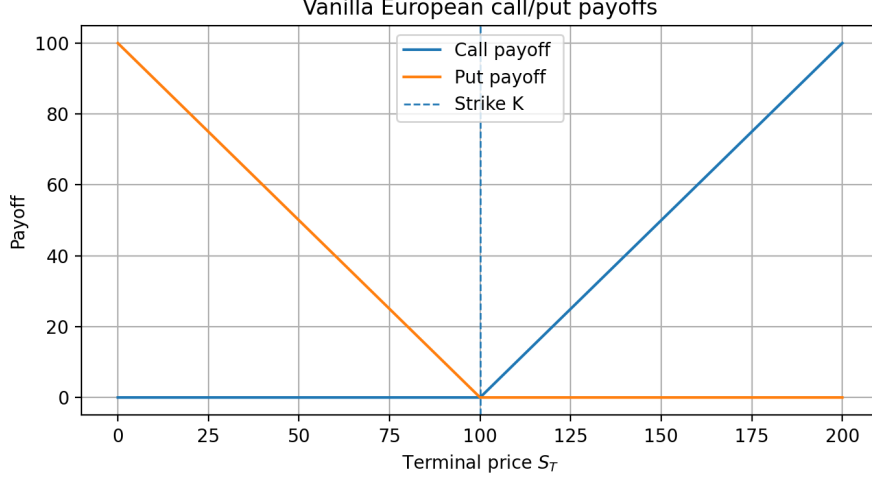


Figure 1: Vanilla European call/put payoffs as functions of S_T .

1.4 Scope of this baseline

Although the framework applies to many derivatives (digitals, barriers, Asians, baskets, best-of, spreads, etc.), we use plain-vanilla European options as a benchmark to compare classical and quantum Monte Carlo. In particular, we focus on payoffs of the form $\Phi = \Phi(S_T)$, which only require terminal simulation.

2 Models: Black–Scholes and Heston

2.1 Black–Scholes (reference model)

Under $\mathbb{Q}$:

$$\frac{dS_t}{S_t} = (r - q) dt + \sigma dW_t^{\mathbb{Q}}, \quad \sigma > 0.$$

The terminal value is lognormal:

$$S_T = S_0 \exp\left((r - q - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z\right), \quad Z \sim \mathcal{N}(0, 1).$$

Black–Scholes provides closed-form prices for European calls/puts and exact terminal simulation.

2.2 Heston stochastic volatility model

2.2.1 Dynamics under $\mathbb{Q}$

$$\begin{aligned} dS_t &= (r - q) S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t^{(2)}, \end{aligned}$$

and

$$dW_t^{(1)} dW_t^{(2)} = \rho dt.$$

Here v_t is the instantaneous variance, κ the mean-reversion speed, θ the long-run variance, ξ the vol-of-vol, and ρ the spot–variance correlation.

2.2.2 Why Monte Carlo is needed

Unlike Black–Scholes, S_T does not generally admit a simple closed-form distribution. Monte Carlo is therefore a natural baseline:

$$V_0 \approx e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{i=1}^{N_{\text{paths}}} \Phi(S_T^{(i)}),$$

with the main numerical challenge being the non-negativity constraint $v_t \geq 0$ when discretizing.

2.2.3 Black–Scholes as a limiting case

Heston approaches Black–Scholes when variance becomes (approximately) constant. A typical regime is $\xi \rightarrow 0$ with $v_0 = \theta = \sigma^2$ (vanishing variance fluctuations).

3 Classical pricing under Heston: Monte Carlo with Euler discretization

3.1 Time grid and discretization principle

We split $[0, T]$ into N_{steps} equally spaced steps:

$$t_n = n\Delta t, \quad n = 0, \dots, N_{\text{steps}}, \quad \Delta t = \frac{T}{N_{\text{steps}}}.$$

Over one step, Brownian increments satisfy $\Delta W_n \sim \mathcal{N}(0, \Delta t)$, hence

$$\Delta W_n \approx \sqrt{\Delta t} Z_n, \quad Z_n \sim \mathcal{N}(0, 1).$$

Smaller Δt reduces discretization bias but increases computational cost.

3.2 Correlated Gaussian increments

At each step, draw $Z_1, Z_\perp \sim \mathcal{N}(0, 1)$ i.i.d. and set

$$Z_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_\perp.$$

Then

$$\Delta W^{(1)} = \sqrt{\Delta t} Z_1, \quad \Delta W^{(2)} = \sqrt{\Delta t} Z_2.$$

3.3 Variance update: full truncation Euler

Define $v_n^+ = \max(v_n, 0)$ and update

$$v_{n+1} = v_n + \kappa(\theta - v_n^+) \Delta t + \xi \sqrt{v_n^+} \sqrt{\Delta t} Z_2, \quad v_{n+1} \leftarrow \max(v_{n+1}, 0).$$

3.4 Spot update: log-Euler

With $v_n^+ = \max(v_n, 0)$:

$$\log S_{n+1} = \log S_n + (r - q - \tfrac{1}{2} v_n^+) \Delta t + \sqrt{v_n^+} \sqrt{\Delta t} Z_1, \quad S_{n+1} = \exp(\log S_{n+1}).$$

3.5 Pricing estimators and error sources

$$\hat{C}_0 = e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{i=1}^{N_{\text{paths}}} (S_T^{(i)} - K)^+, \quad \hat{P}_0 = e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{i=1}^{N_{\text{paths}}} (K - S_T^{(i)})^+.$$

Statistical error decreases as $O(N_{\text{paths}}^{-1/2})$ while decreasing Δt reduces discretization bias:

$$\text{Total error} \approx \underbrace{\text{discretization bias}}_{\Delta t \rightarrow 0} + \underbrace{\text{statistical error}}_{O(N_{\text{paths}}^{-1/2})}.$$

3.6 Numerical parameters used for the figures

All figures in the remainder of this report that involve Heston Monte Carlo simulations (price surface, implied-volatility surface, smiles, and Monte Carlo diagnostics) are generated using the following baseline parameter set.

Market inputs.

$$S_0 = 100, \quad r = 0.02, \quad q = 0.00.$$

Heston parameters (synthetic baseline).

$$\kappa = 2.0, \quad \xi = 0.5, \quad \rho = -0.7, \quad v_0 = 0.04, \quad \theta = 0.04.$$

The negative spot–variance correlation $\rho < 0$ produces a typical equity-style implied-volatility skew.

Monte Carlo configuration. We use

$$N_{\text{steps}} = 200, \quad N_{\text{paths}} = 10,000,$$

with the full truncation Euler scheme for v_t and a log-Euler update for S_t as described above.

Grid of strikes and maturities.

$$K \in \{70, 75, 80, 85, 90, 95, 100, 105, 110\}, \quad T \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0\}.$$

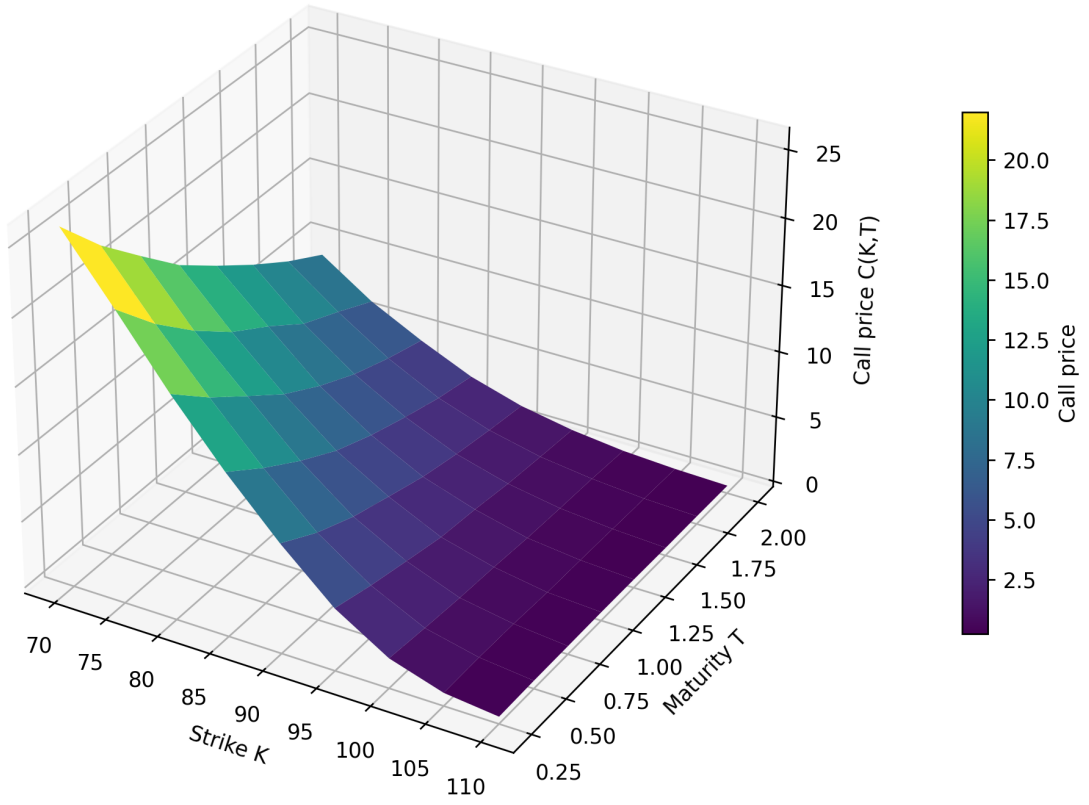


Figure 2: Monte Carlo Heston call price surface $C(K, T)$ (synthetic example).

3.7 From prices to implied volatility

Markets often quote options in implied volatility. We define $\sigma_{\text{imp}}(K, T)$ by

$$C^{BS}(\sigma_{\text{imp}}(K, T)) = C(K, T).$$

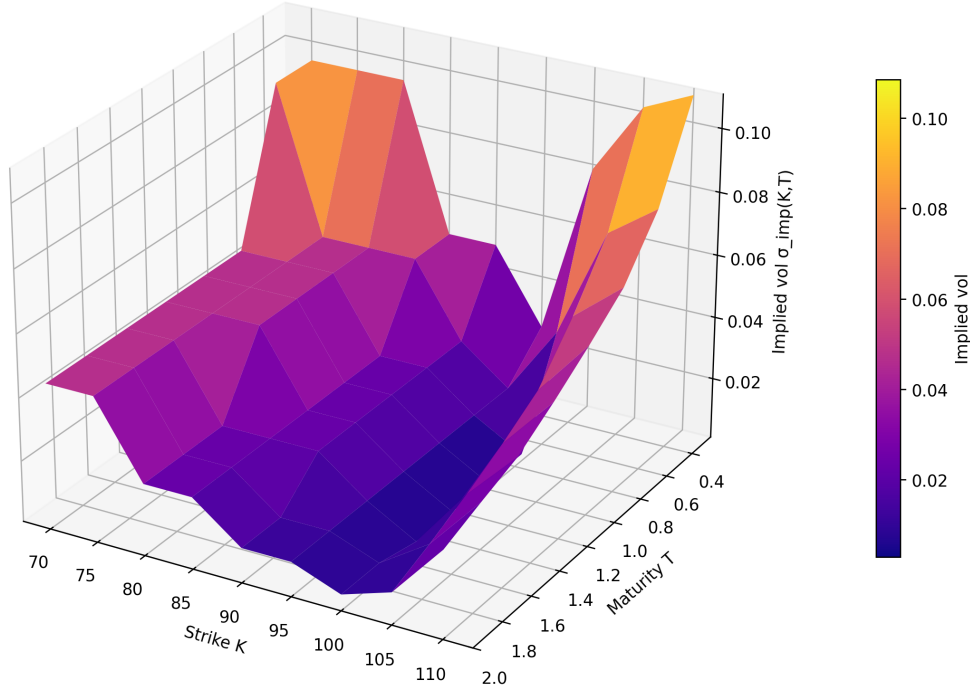


Figure 3: Implied volatility surface $\sigma_{\text{imp}}(K, T)$ obtained by Black–Scholes inversion (synthetic example).

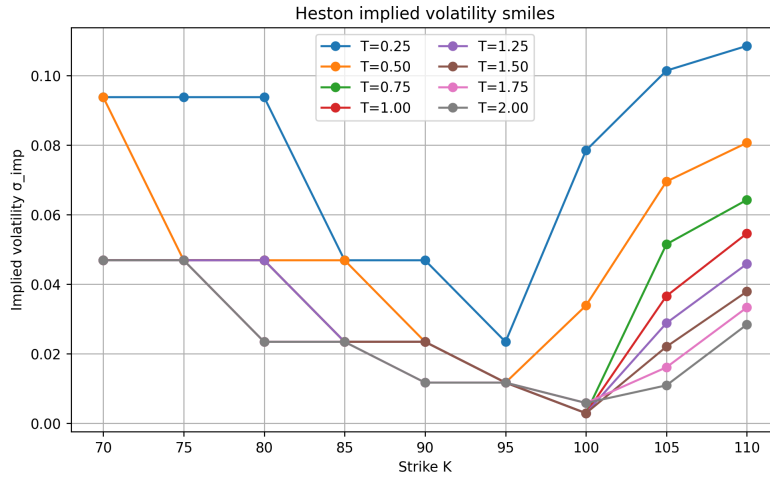


Figure 4: Implied volatility smiles $\sigma_{\text{imp}}(K)$ for several maturities (synthetic example).

4 Variance reduction and Monte Carlo diagnostics

4.1 Antithetic variates (general case)

Setting. Let $Z \in \mathbb{R}^d$ be a random vector with a symmetric distribution (e.g. $Z \sim \mathcal{N}(0, \Sigma)$) and let $Y = g(Z)$ be the quantity of interest (e.g. discounted payoff).

Antithetic construction. For each draw Z , also use $-Z$ and average:

$$Y^{\text{anti}}(Z) = \frac{1}{2} \left(g(Z) + g(-Z) \right).$$

Because $Z \stackrel{d}{=} -Z$, the estimator is unbiased:

$$\mathbb{E}[Y^{\text{anti}}(Z)] = \mathbb{E}[g(Z)].$$

Why variance can decrease. Let $Y_1 = g(Z)$ and $Y_2 = g(-Z)$. Then

$$\text{Var}\left(\frac{Y_1 + Y_2}{2}\right) = \frac{1}{2} \text{Var}(Y) + \frac{1}{2} \text{Cov}(g(Z), g(-Z)).$$

If $\text{Cov}(g(Z), g(-Z)) < 0$, then $\text{Var}(Y^{\text{anti}}) < \text{Var}(Y)$.

4.2 Empirical illustration: variance reduction

We compare plain Monte Carlo and antithetic variates using the 95% confidence interval width (mean over repeated runs).

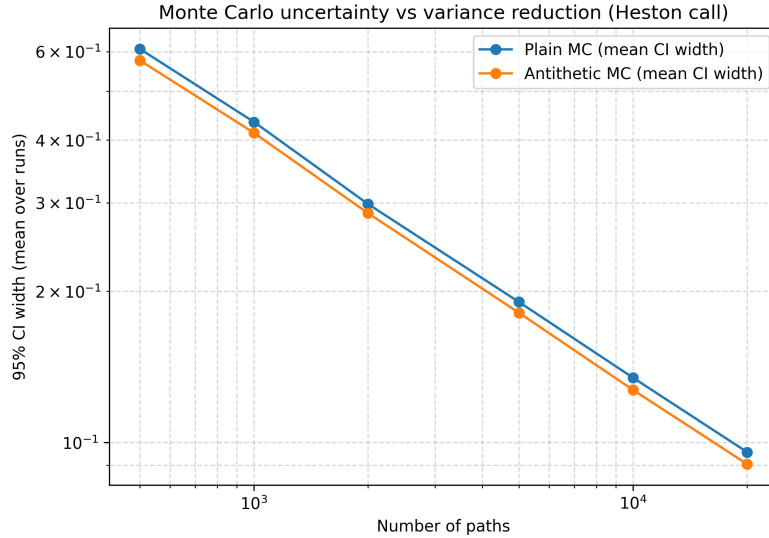


Figure 5: Monte Carlo uncertainty vs variance reduction (Heston call): 95% CI width vs number of paths.

5 Heston calibration to market implied volatility

5.1 Method

In practice, Heston parameters are not known and must be inferred from market option quotes. We calibrate the risk-neutral parameter vector

$$\Theta = (\kappa, \theta, \xi, \rho, v_0)$$

to implied volatilities observed on the SPY (S&P 500 ETF) option surface.

Market dataset. Each option quote provides $(K, T, \sigma_{\text{mkt}}(K, T))$. To reduce microstructure noise and illiquid tails, we discard very short maturities, remove strikes far from spot, and filter out inconsistent implied volatility values.

Calibration objective. To avoid repeatedly inverting Black–Scholes inside the optimizer, we first convert market implied volatilities into Black–Scholes prices:

$$C_{\text{mkt}}(K, T) = C^{BS}(S_0, K, T, r, q, \sigma_{\text{mkt}}(K, T)).$$

Given Θ , we compute the Heston model price $C_H(K, T; \Theta)$ using characteristic-function pricing, and minimize a mean-squared error in prices:

$$\mathcal{L}(\Theta) = \frac{1}{N} \sum_{(K, T)} \left(C_H(K, T; \Theta) - C_{\text{mkt}}(K, T) \right)^2.$$

Constraints and optimization. We enforce

$$\kappa > 0, \theta > 0, \xi > 0, v_0 > 0, -1 < \rho < 1,$$

and solve the bounded problem with L-BFGS-B from a reasonable initial guess.

5.2 Calibrated parameters (SPY)

Table 1 reports the calibrated parameters obtained on the SPY option surface.

Parameter	κ	θ	ξ	ρ	v_0
Calibrated value	1.986341	0.076097	0.709019	-0.959721	0.028956

Table 1: Calibrated risk-neutral Heston parameters for SPY (S&P 500 ETF).

5.3 Post-calibration diagnostics

After calibration, we compute the implied volatility produced by the calibrated Heston model:

$$\sigma_H(K, T) \text{ such that } C^{BS}(S_0, K, T, r, q, \sigma_H(K, T)) = C_H(K, T; \hat{\Theta}),$$

and compare σ_H to σ_{mkt} on selected maturities. Figure 6 shows implied-volatility smiles (IV vs strike K), with maturities T in the legend.

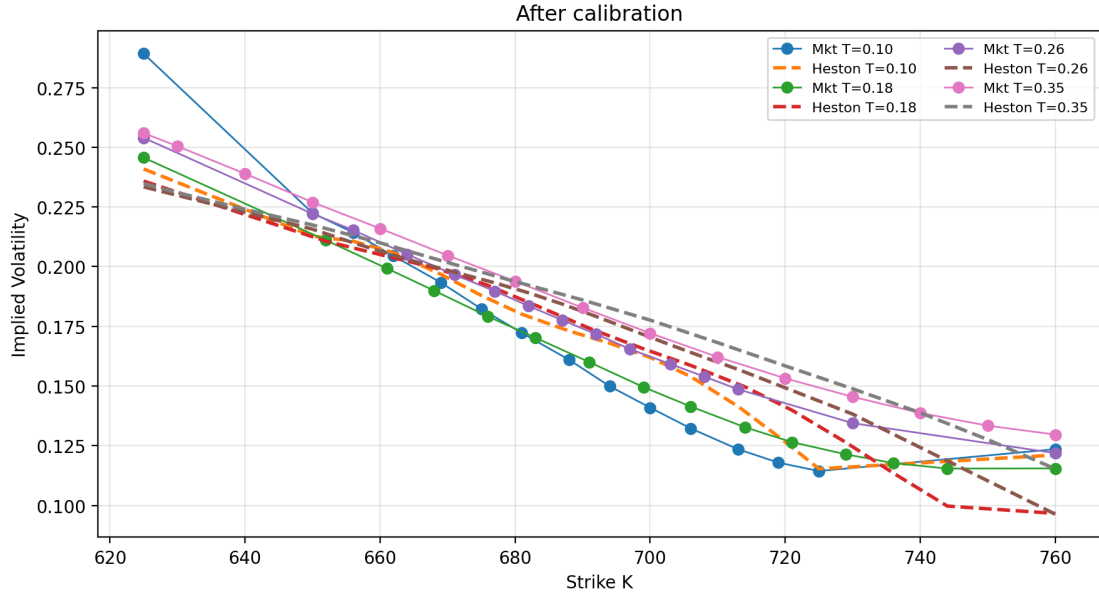


Figure 6: After calibration (Heston): market implied volatilities vs calibrated Heston implied volatilities as a function of strike K , for selected maturities T (legend).

5.4 Conclusion of the classical baseline

- Black–Scholes provides a clean reference (closed-form vanillas, exact terminal simulation).
- Heston introduces stochastic volatility and reproduces implied-volatility skew/smile effects.
- Monte Carlo is universal but converges slowly; Heston also introduces discretization bias.
- Variance reduction improves the classical baseline and is essential for fair comparison with quantum methods.

6 Quantum Monte Carlo and Amplitude Estimation

6.1 From Classical Monte Carlo to Quantum Monte Carlo

Classical Monte Carlo estimates an expectation by drawing N independent samples and averaging them. The statistical error decreases as

$$|\hat{\mu} - \mu| = O(N^{-1/2}).$$

Quantum Monte Carlo targets the same expectations, but first encodes them as a single measurement probability $\mu = \Pr(\text{ancilla} = 1)$. In the ideal case, one can achieve

$$|\hat{\mu} - \mu| = O(N^{-1}),$$

leading to a quadratic speedup in terms of oracle calls.

6.2 Quantum Amplitude Estimation

6.2.1 Definition

QAE estimates a single probability $\mu \in [0, 1]$ by rewriting the target expectation as a probability.

6.2.2 Step 1: Rewrite a generic target as a weighted sum

$$\mu = \sum_{i=0}^{N-1} p_i f_i, \quad \sum_{i=0}^{N-1} p_i = 1, \quad f_i \in [0, 1]. \quad (3)$$

6.2.3 Step 2: encoding μ as a measurement probability (vector view)

Define

$$x = \begin{pmatrix} \sqrt{p_0} \sqrt{1 - f_0} \\ \vdots \\ \sqrt{p_{N-1}} \sqrt{1 - f_{N-1}} \\ \sqrt{p_0} \sqrt{f_0} \\ \vdots \\ \sqrt{p_{N-1}} \sqrt{f_{N-1}} \end{pmatrix}, \quad \Pr(\text{ancilla} = 1) = \sum_{i=0}^{N-1} p_i f_i = \mu.$$

6.2.4 Sampling Monte Carlo on the ancilla

$$\hat{\mu}_{\text{sampling}} = \frac{1}{\text{shots}} \sum_{m=1}^{\text{shots}} \mathbf{1}\{\text{ancilla}^{(m)} = 1\}, \quad |\hat{\mu} - \mu| = O(\text{shots}^{-1/2}).$$

6.2.5 QAE and Grover amplification

Geometric intuition: Bloch sphere and rotations.

$$|\psi\rangle = \cos\left(\frac{\vartheta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\vartheta}{2}\right) |1\rangle. \quad (4)$$

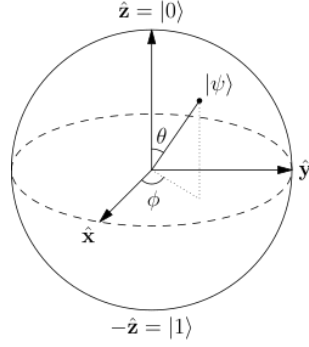


Figure 7: Bloch sphere representation of a qubit state (angles ϑ and φ).

Grover rotation in a 2D subspace. Let

$$v_0 = \begin{pmatrix} \sqrt{1-\mu} \\ \sqrt{\mu} \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \mu = \sin^2 \theta.$$

After k Grover iterations,

$$\Pr(\text{ancilla} = 1 \text{ after } Q^k) = \sin^2((2k+1)\theta).$$

6.2.6 Application: European option pricing as amplitude estimation

Discretization (NISQ constraint). Using n state qubits, $N = 2^n$ bins give (p_i, S_i) and

$$\mathbb{E}^{\mathbb{Q}}[\Phi(S_T)] \approx \sum_{i=0}^{N-1} p_i \Phi(S_i).$$

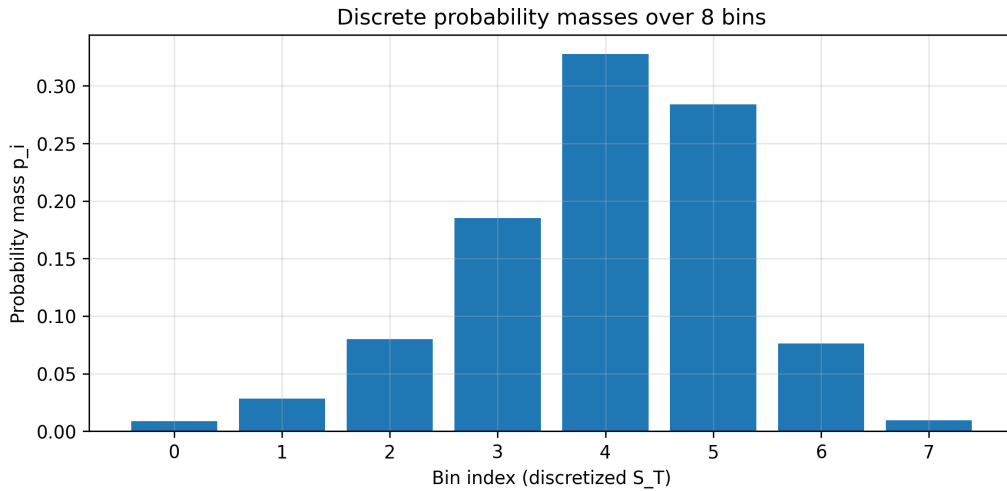


Figure 8: Discretization of S_T : classical Monte Carlo sample $\rightarrow$ histogram bins $\rightarrow (p_i, \Phi(S_i))$ for $N = 2^n$ (placeholder).

Normalization. Let $g_i = \Phi(S_i)$ and $f_i = g_i/g_{\max} \in [0, 1]$. Then $V_0^{\text{disc}} = e^{-rT} g_{\max} \mu$.

6.2.7 Error decomposition and NISQ limitations

$$|V_0 - \widehat{V}_0| \leq |V_0 - V_0^{\text{disc}}| + |V_0^{\text{disc}} - \widehat{V}_0^{\text{disc}}| + \text{hardware / implementation error}.$$

7 Implementation and experiments: Qiskit ancilla sampling

In practice, most of our numerical experiments rely on **ancilla sampling** (i.e., repeated measurements of the objective qubit) rather than full QAE. This choice is motivated by (i) NISQ depth constraints and (ii) software/hardware compatibility issues when running iterative amplitude estimation end-to-end. The resulting estimator has the same $O(\text{shots}^{-1/2})$ statistical scaling as classical Monte Carlo, but it preserves the *quantum encoding* pipeline (state preparation and payoff-to-amplitude mapping), which is the key proof-of-concept of this work.

7.1 Quantum encoding pipeline (state preparation + payoff encoding)

For each pair (K, T) , we build a quantum circuit $A_{K,T}$ composed of two blocks:

1. **Uncertainty model (state preparation).** Using n state qubits, we discretize the terminal price S_T on a grid $\{S_i\}_{i=0}^{2^n-1}$ with associated probabilities $\{p_i\}$. The circuit prepares the superposition

$$\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle,$$

where the computational basis state $|i\rangle$ encodes the bin S_i .

2. **Payoff encoding (objective qubit).** We encode the (normalized) payoff into a dedicated objective qubit via controlled rotations such that, conditionally on S_i ,

$$|i\rangle|0\rangle \mapsto |i\rangle \left(\sqrt{1-a(S_i)}|0\rangle + \sqrt{a(S_i)}|1\rangle \right),$$

hence

$$\Pr(1 | S_i) = a(S_i).$$

After both blocks, the probability of observing 1 on the objective qubit equals

$$a = \Pr(1) = \sum_{i=0}^{2^n-1} p_i a(S_i) \approx \mathbb{E}[\widetilde{\Phi}(S_T)],$$

where $\widetilde{\Phi}$ is a normalized version of the payoff (so that it can be represented as a probability in $[0, 1]$).

7.2 Ancilla sampling estimator

We estimate the probability $a = \Pr(1)$ by repeatedly measuring the objective qubit. Let $C^{(m)} \in \{0, 1\}$ be the measurement outcome of the objective qubit at shot m . With $\text{shots} = M$ circuit repetitions,

$$\hat{a} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{C^{(m)} = 1\}, \quad \text{Std}(\hat{a}) \approx \sqrt{\frac{a(1-a)}{M}}.$$

The option value is then recovered using the deterministic post-processing map provided by the payoff-encoding circuit:

$$\mathbb{E}[\widehat{\Phi(S_T)}] = \text{post_processing}(\hat{a}), \quad \widehat{V}_0 = e^{-rT} \text{post_processing}(\hat{a}).$$

In our Qiskit implementation, this corresponds to:

- creating a copy of the circuit and adding a 1-bit classical register,
- measuring the objective qubit into this classical bit,
- running the circuit for M shots and extracting $\hat{a} = \Pr(1)$,
- applying `post_processing` and discounting by e^{-rT} .

7.3 Discretized reference (“Exact”) for validation

To validate the quantum sampling estimate, we compute a discretized reference on the same grid:

$$V_0^{\text{disc}}(K, T) = e^{-rT} \sum_{i=0}^{2^n-1} p_i \Phi(S_i), \quad \Phi(S) = \max(S - K, 0).$$

This “Exact” quantity is exact for the *discretized and truncated* model used to prepare the quantum state.

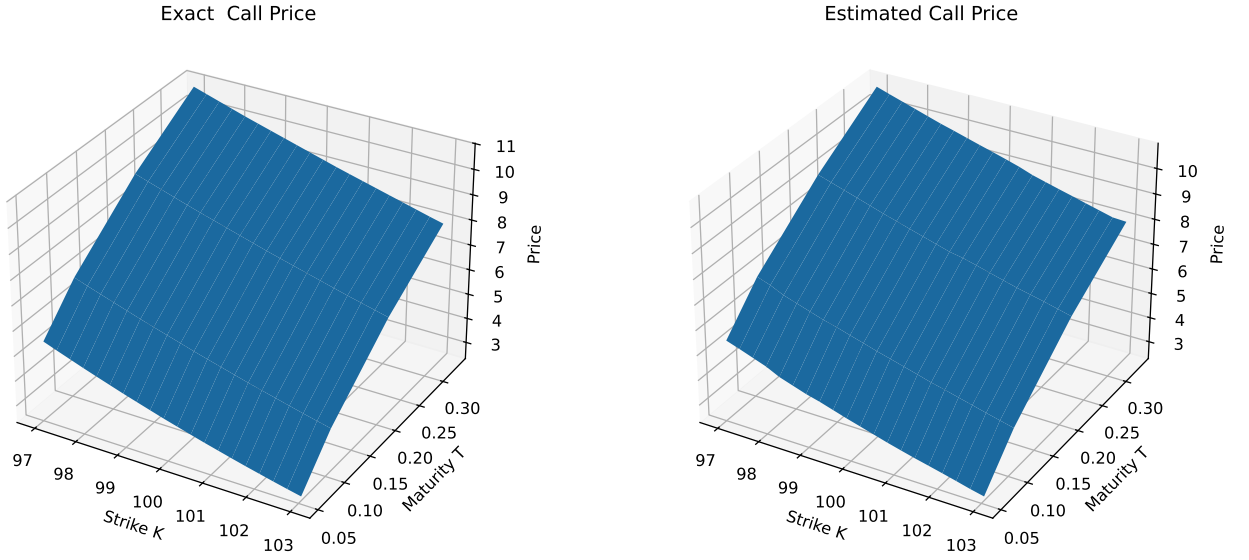


Figure 9: Call price surfaces: exact (left), estimated (right).

7.4 Error reporting

We report the signed error surface

$$\text{Error}(K, T) = \hat{V}_0^{\text{disc}}(K, T) - V_0^{\text{disc}}(K, T),$$

as well as summary statistics (maximum absolute error, mean absolute error) over the (K, T) grid.

Error (Estimated - Exact)

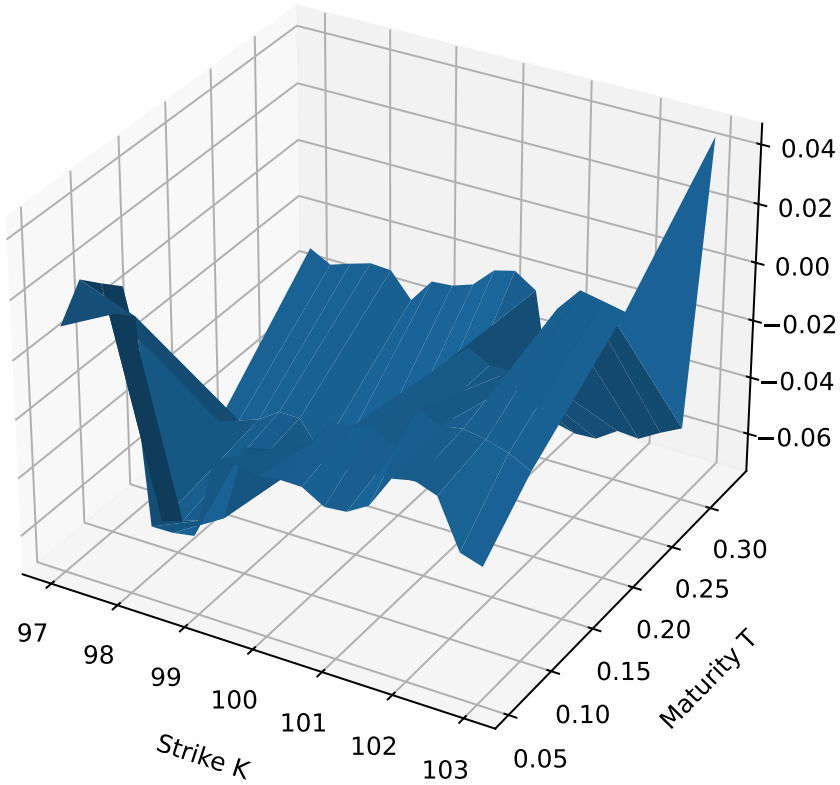


Figure 10: Call price surfaces: Error.

7.5 Experiments on a real 3-qubit quantum device (NMR)

In this project, a subset of experiments is executed on a **real 3-qubit Nuclear Magnetic Resonance (NMR)** quantum processor (SpinQ Triangulum). Qubits are implemented as nuclear spin- $\frac{1}{2}$ degrees of freedom in a liquid-state molecular ensemble. The sample contains iodotrifluoroethylene ($\text{C}_2\text{F}_3\text{I}$), whose three ^{19}F nuclei act as the three qubits. Quantum gates are realized by radio-frequency (RF) control pulses, and measurements correspond to ensemble-averaged readout signals.

Due to hardware constraints (highly mixed initial state, ensemble readout, and limited circuit depth), these runs are intended as a **proof-of-concept** validation of shallow state preparation and payoff encoding, followed by ancilla sampling (rather than deep QAE circuits).

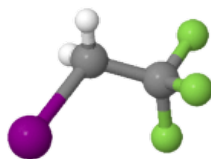


Figure 11: Iodotrifluoroethylene ($\text{C}_2\text{F}_3\text{I}$) used in the liquid-state NMR sample. The three ^{19}F nuclei (green) serve as the three qubits.

7.6 Experimental setup (QPU runs)

- **Registers:** $n = 2$ state qubits + 1 objective qubit (total: 3 qubits).
- **Target quantity:** $a = \Pr(1)$ on the objective qubit, mapped to $\hat{V}_0$ via post-processing.
- **Shots:** we report the number of repetitions per circuit.
- **Compilation/transpilation:** we use a shallow circuit layout compatible with the device constraints.